EPSR-test-ex58 local_density: multi-bond cluster analysis

Program(s): local_density | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises `local_density`, which despite its name is the multiple-bond variant of `clusters` [1]: it uses the `.CLUSTERS.dat` keyword table, but `atom1`, `atom2`, `rmin` and `rmax` may be lists, so that two molecules are bonded if any listed atom pair lies within its range. Connected molecules form clusters, whose size distribution, percolating fraction and fraction in clusters of two or more molecules are written.

Code exercised

Fortran (original, fortran_src/):

- `local_density.f`: `run_clusters` (word lists, column titles), `find_clusters` (bond test, cluster labelling, percolation)
- `parse.f`: `parse` (word lists of `atom1`, `atom2`, `rmin`, `rmax`)
- `read_and_save_files.f`: `outgofz`, `save_files_others`; `setup_symbols_generic.f`: `setup_symbols_clusters`

Rust (port, rust/epsr-core/src/):

- `tools/local_density.rs`: `run`, `find_clusters`
- `fortran/text.rs`: `parse`, `parse_n`; `tools/anal_common.rs`: `outgofz`, `write_fort14`
- `symbols.rs`: `setup_input_file`, `read_input_file`, `save_inp_file`; `output.rs`: `r_int`; `symtab_tools.rs`: `CLUSTERS`

Physical system

Analysed configuration: `200DMSO400Water.ato`, an unchanged copy of the EPSR-test box of 200 dimethyl sulphoxide (DMSO; atom types S, O, C, H) and 400 water molecules (O1, H1): 3200 atoms, DMSO mole fraction 1/3, 300 K, cubic box of side 33.1213 Å (0.0881 atoms/Å^3), the EPSR model of the deuterated DMSO-water neutron data (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), cf. [2]. Cluster 1: the water hydrogen-bond network (O1···H1 1.2-2.4 Å); cluster 2: water-water plus DMSO O···water H1 (1.2-2.5 Å) hydrogen bonds; cluster 3: mixed clusters from S-S 0-5 Å, C···O 2.0-3.2 Å and O1···H1 1.5-2.0 Å contacts. The reference `fort.14` gives fractions in clusters of 2 or more molecules of 0.975, 0.965 and 0.888.

Inputs

`hbnet.CLUSTERS.dat`: `nsize 700`, `nsumt 0`, three clusters with one, two and three bond definitions. `200DMSO400Water.ato` copied unchanged from EPSR-test; the keyword file was written by hand.

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
`repeat 2` with `nsumt 0`, so pass 2 averages with pass 1. Checks the parsing of the word lists, the composed column titles (`O1-H1, O-H1`, `S-S, C-O, O1-H1` in the reference `.n01`), the cluster labelling, the percolating fraction (row 1) and the fraction-in-clusters lines. Compared after each pass: `hbnet.CLUSTERS.n01`, `hbnet.CLUSTERS.dat`, `fort.14`. The configuration is the same in every pass, so the chained passes test the read-back of the accumulated file and the running-average arithmetic.

Notes

`fort.14`: the Fortran writes these lines to unit 14 after `save_files_others` has closed it, so gfortran creates the default file `fort.14`; the port writes the same file.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[2] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[3] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.